

Nova Bank - Feature Documentation

A Premium Digital Banking Platform Built for Sri Lanka

Full-Stack • Next.js + NestJS • TypeORM + SQLite • Real-Time SSE • AI-Powered Insights

Created by team **We Forgot the Semicolon** in the Hack 2 Night event.

Tech Stack Overview

Layer	Technology
Frontend	Next.js 15 (App Router), React 18, CSS Modules
Backend	NestJS 10, TypeORM, class-validator
Database	SQLite (file-based, zero-config)
Auth	JWT (Bearer Token), bcrypt password hashing
Real-Time	Server-Sent Events (SSE)
AI	Google Gemini API (financial advisor)
Email	Nodemailer (SMTP or local file inspector)
File Storage	AWS S3 (avatar uploads)
API Docs	Swagger / OpenAPI auto-generated

1. Authentication & Security

What it does: Secure user registration, login, password reset with email verification codes, and JWT-based session management.

Key Capabilities

- User registration with username, full name, email, and password
- Login returns JWT token stored in localStorage
- Password reset via 6-digit email verification codes (OTP)
- Role-based access control (customer vs admin)
- JWT guard protects all authenticated endpoints

File Locations

Component	File Path
Login Page	app/(accounts)/login/page.tsx
Sign Up Page	app/(accounts)/sign-up/page.tsx
Reset Password Page	app/(accounts)/reset-password/page.tsx
Auth Controller	server/src/auth/auth.controller.ts
Auth Service	server/src/auth/auth.service.ts
JWT Guard	server/src/common/guards/jwt-auth.guard.ts
Roles Guard	server/src/common/guards/roles.guard.ts

2. Dashboard

What it does: The central command center showing account balances, recent transactions, quick actions, and bill splitting functionality.

Key Capabilities

- View all linked bank accounts with real-time balances
- Switch between accounts to see per-account transactions
- Quick access to recent transaction history
- One-click bill splitting directly from transaction rows
- Saved payees quick access panel
- Responsive layout with sidebar navigation

File Locations

Component	File Path
Dashboard Page	app/dashboard/page.tsx

3. Bank Accounts Management

What it does: Full CRUD management for linked bank accounts with add, edit (nickname), and viewing capabilities.

Key Capabilities

- View all accounts with balances in a clean card layout
- Add new bank accounts (account number, name, email)
- Edit account nicknames
- Search/filter accounts
- URL-driven mode switching (?mode=edit, ?mode=add)

File Locations

Component	File Path
Accounts Page	app/bank-accounts/page.tsx
Accounts CSS	app/bank-accounts/accounts.module.css
Accounts Controller	server/src/accounts/accounts.controller.ts
Accounts Service	server/src/accounts/accounts.service.ts

4. Virtual Card Management

What it does: A premium card dashboard where users generate virtual debit/credit cards for online shopping with interactive 3D card mockups.

Key Capabilities

- **Generate** new virtual debit or credit cards linked to any account

- **Interactive card flip** animation - click to reveal CVV on the back
- **Freeze / Unfreeze** toggle to dynamically block/allow transactions
- **Daily spending limit** slider per card
- Premium card design with chip, embossed numbers, and Nova branding

UI Highlights

- 3D CSS card flip with perspective and backface-visibility
- Frosted glass overlay when card is frozen
- Smooth slider for spending cap adjustment

File Locations

Component	File Path
Cards Page	app/cards/page.tsx
Cards CSS	app/cards/cards.module.css
Virtual Cards Controller	server/src/virtual-cards/virtual-cards.controller.ts
Virtual Cards Service	server/src/virtual-cards/virtual-cards.service.ts

5. Bank Transfers

What it does: Full-featured money transfer system supporting instant transfers, scheduled/recurring payments, virtual card payments, and saved payees.

Key Capabilities

- **Instant transfers** - account-to-account with real-time balance deduction
- **Scheduled transfers** - recurring (daily, weekly, monthly) transfers with cron-based execution
- **Virtual card payments** - pay using any linked virtual card (enforces freeze check + daily limit)
- **Saved payees** - save frequently used beneficiaries for quick access
- **Multi-step confirmation** - Form → Review → Success/Failure screens
- **Pending bill split requests** - view and approve/decline split requests from friends

File Locations

Component	File Path
Transfer Page	app/bank-transfer/page.tsx
Transfer Controller	server/src/transfer/transfer.controller.ts
Transfer Service	server/src/transfer/transfer.service.ts
Scheduled Transfers	server/src/scheduled-transfers/
Payees Controller	server/src/payees/

6. Bill Payments

What it does: Pay utility bills to 12+ Sri Lankan service providers (CEB, Dialog, Airtel, SLT, Water Board, HSBC, AIA, etc.).

Key Capabilities

- Grid of biller logos with search filtering
- Select biller → enter bill reference + amount + source account
- Multi-step flow: Select Biller → Fill Form → Success/Failure
- Animated success/failure screens
- Card payment support (pay bills using virtual cards)

File Locations

Component	File Path
Pay Bills Page	app/pay-bills/page.tsx

7. Bill Splitting with Friends (Social Finance)

What it does: Split any past transaction with another Nova Bank customer via username search.

Key Capabilities

- Select a transaction from the dashboard (e.g., "Dinner Cafe Noir - Rs. 3,500")
- Search for friends by username and set a split amount
- Pushes a **real-time SSE notification** to the friend: *"Dilara requested Rs. 1,750 for Lunch"*
- Friend can **approve** (auto-deducts and transfers) or **decline**
- Approval triggers counter-notification to the requester

File Locations

Component	File Path
Split Modal (Dashboard)	app/dashboard/page.tsx
Pending Splits (Transfer)	app/bank-transfer/page.tsx
Bill Splits Controller	server/src/bill-splits/

8. Smart Spend - AI-Powered Budget Analytics

What it does: A rich analytics dashboard with spending breakdowns, trend charts, budget management, and AI-generated financial insights.

Key Capabilities

- **Spending Summary** - interactive donut chart by category (Food, Bills, Shopping, Entertainment, Transport, Others)
- **6-Month Trend** - bar chart showing monthly spend evolution
- **Budget Manager** - set spending limits per category, visual progress bars (green → amber → red)
- **AI Insights** - Gemini-powered one-liner recommendations (e.g., *"Your food spend is 40% above last month. Consider meal prepping."*)
- Create, edit, and delete budgets with category-based controls
- Budget exceeded notifications via SSE

File Locations

Component	File Path
Smart Spend Page	app/smart-spend/page.tsx
Insights Controller	server/src/insights/insights.controller.ts
Insights Service	server/src/insights/insights.service.ts

Component	File Path
Budgets Module	server/src/budgets/

9. Nova AI - Financial Advisor Chatbot

What it does: A floating AI chat assistant powered by Google Gemini that answers questions about your finances.

Key Capabilities

- Floating chat bubble (bottom-right) with slide-up animation
- Context-aware: the AI sees your accounts, recent transactions, and budget status
- Natural language Q&A: *"How much did I spend on food this month?"*, *"Am I over budget?"*
- Streaming-like response display
- Personalized greeting with user's full name

File Locations

Component	File Path
AI Chat Component	components/ai-chat.tsx
AI Controller	server/src/ai/ai.controller.ts
AI Service	server/src/ai/ai.service.ts

10. E-Statements

What it does: Generate detailed bank statements for any account with running balance calculations.

Key Capabilities

- Select account → auto-generates tabular statement
- Each row shows: Date, Description, Debit, Credit, Running Balance
- Summary card: Opening Balance, Total Credits, Total Debits, Closing Balance
- Debit/credit color-coded rows

File Locations

Component	File Path
E-Statement Page	app/e-statement/page.tsx
Statements Module	server/src/statements/

11. Interactive Savings Jars & Spare Change Round-Ups

What it does: A gamified savings goals dashboard with 3D glass jar visualizations and automatic spare change collection.

Key Capabilities

- **Create savings jars** with name, target amount, and optional icon
- **3D Glass Jar UI** - transparent containers with CSS reflections, glossy highlights, and lid/neck details
- **Gold liquid fill** - dynamic height proportional to $\text{currentAmount} / \text{targetAmount}$, with animated wave overlays and floating bubble particles
- **Manual deposit/withdraw** - move funds between accounts and jars via modal forms
- **Spare Change Round-Up** - when enabled, every card transaction is automatically rounded up to the nearest Rs. 50 or Rs. 100, and the difference is swept into the active jar
- **Smart safeguards** - skips round-up if insufficient balance; sends notification either way
- **One active jar** - only one jar can have round-up enabled at a time

File Locations

Component	File Path
Savings Jars Page	app/savings-jars/page.tsx
Savings Jars CSS	app/savings-jars/savings-jars.module.css
Savings Jars Entity	server/src/savings-jars/entities/savings-jar.entity.ts
Savings Jars Service	server/src/savings-jars/savings-jars.service.ts
Savings Jars Controller	server/src/savings-jars/savings-jars.controller.ts
Round-Up Logic	server/src/transfer/transfer.service.ts

12. Real-Time Notification System (SSE)

What it does: A live notification center with Server-Sent Events that pushes instant alerts across the platform.

Key Capabilities

- **SSE stream** - persistent connection to /notifications/stream for real-time delivery
- **Bell icon** with unread badge counter (9+)
- **Dropdown panel** - list all notifications, color-coded dots by type
- **Toast alerts** - premium slide-in banner for new notifications (auto-dismiss after 5s)
- **Mark as read** - single or mark-all-read
- Triggered by: transfers, bill splits, budget alerts, round-up sweeps, and more

File Locations

Component	File Path
Notification Center	components/notification-center.tsx
Notifications Module	server/src/notifications/

13. User Profile Management

What it does: View and edit personal profile details including avatar, name, email, and NIC.

Key Capabilities

- **Avatar upload** - click-to-upload with drag overlay, uploaded to AWS S3
- **Glowing ring** - gradient-bordered circular avatar with hover camera overlay
- **Profile form** - edit Full Name, Email, NIC
- **Role badge** - displays user role (CUSTOMER / ADMIN)
- **Account username** - read-only locked field

File Locations

Component	File Path
Profile Page	app/profile/page.tsx
Users Controller	server/src/users/

14. Email System

What it does: Sends transactional emails via SMTP or logs them locally for development.

Key Capabilities

- **Dual mode:** Real SMTP delivery (production) or local file inspector (dev)
- **HTML templates** - styled email bodies with Nova Bank branding
- **Verification codes** - 6-digit OTP for password reset flow

File Locations

Component	File Path
Mail Service	server/src/mail/mail.service.ts

15. Admin Panel (Control Center)

What it does: A premium administrative command center accessible exclusively to admin accounts.

Key Capabilities

- **Platform Overview Stats (KPIs):** Total Platform Users count, Linked Accounts count, Vault Reserves, and Platform Transaction Volume.
- **Access Denied Gate:** If a regular user attempts to access /admin, an elegant "Access Denied" warning card blocks navigation.
- **User Directory:** Searchable list of users. Role badge toggling.
- **Account Vaults:** Searchable accounts list. Balance display with quick action "Adjust Balance".
- **Balance Adjustments:** Override balance via Deposit, Withdraw, or Set explicit balance forms.
- **System Audit Logs:** Scrollable feed of all platform-wide system audit events.
- **Security Check:** Enforces standard JWT and role checks on backend routes.

File Locations

Component	File Path
Admin Page	app/admin/page.tsx

Component	File Path
Admin CSS	app/admin/admin.module.css
Admin Controller	server/src/admin/admin.controller.ts
Admin Service	server/src/admin/admin.service.ts

Shared Components & Infrastructure

Component	File Path	Purpose
Sidebar	components/sidebar.tsx	Navigation with 8 menu items + settings/help footer
Icons	components/icons.tsx	Custom SVG icon library
Auth Context	lib/auth-context.tsx	React context for global auth state
API Client	lib/api-client.ts	Typed fetch wrapper with JWT header injection
Global CSS	app/globals.css	Design tokens, typography, base styles
App Module	server/src/app.module.ts	Root NestJS module wiring all features
Main Entry	server/src/main.ts	Server bootstrap with CORS, Swagger, and validation



Project Structure

nova-bank/

- |— app/ # Next.js Frontend Pages
 - | |— (accounts)/ # Auth pages (login, sign-up, reset-password)
 - | |— dashboard/ # Main dashboard
 - | |— bank-accounts/ # Account management
 - | |— cards/ # Virtual card management
 - | |— savings-jars/ # Savings goals + round-ups
 - | |— bank-transfer/ # Money transfers
 - | |— pay-bills/ # Utility bill payments
 - | |— smart-spend/ # AI analytics & budgets
 - | |— e-statement/ # Bank statements
 - | |— profile/ # User profile
- |— components/ # Shared React Components
 - | |— sidebar.tsx # Navigation sidebar
 - | |— ai-chat.tsx # Floating AI chatbot
 - | |— notification-center.tsx # SSE notification center
 - | |— Icons.tsx # Custom SVG icons
- |— lib/ # Frontend Utilities
 - | |— auth-context.tsx # Auth state management
 - | |— api-client.ts # API fetch wrapper
- |— server/src/ # NestJS Backend
 - | |— auth/ # Authentication (JWT, register, login)
 - | |— users/ # User profile & avatar
 - | |— accounts/ # Bank account CRUD
 - | |— transactions/ # Transaction records
 - | |— transfer/ # Transfer execution + round-up
 - | |— virtual-cards/ # Card generation & management
 - | |— savings-jars/ # Savings goals CRUD
 - | |— bill-splits/ # Social bill splitting
 - | |— budgets/ # Category budgets
 - | |— insights/ # Spending analytics
 - | |— ai/ # Gemini AI chatbot
 - | |— notifications/ # SSE + notification storage
 - | |— scheduled-transfers/ # Cron-based recurring payments

— payees/	# Saved beneficiaries
— statements/	# E-statement generation
— mail/	# Email service (SMTP/local)
— admin/	# Admin system stats
— search/	# Global search
— health/	# Health check endpoint

Design Philosophy

- **Premium Dark Sidebar** - deep purple (#450043) with rounded corners and active state highlighting
- **Glassmorphism** - frosted glass cards with backdrop-filter: blur() and subtle borders
- **Micro-Animations** - card flips, wave effects, bubble particles, slide-in toasts, progress fills
- **Color-Coded Categories** - consistent palette across charts, budgets, and notifications
- **Mobile Responsive** - all pages adapt via CSS media queries at 768px breakpoint